Shenzhen Hailingke Electronics Co., Ltd.

HLK-LD2420 User Manual

Table of Contents

1. Product Information

HLK-LD2420 is a high-performance 24 GHz radar module with a transmit-receive antenna from Hilink. Its human body sensing algorithm uses millimeter wave

Radar distance measurement technology and the S3 series chip's advanced proprietary signal processing technology enable accurate perception of motion, micro-movement and standing human bodies.

HLK-LD2420 is mainly used in indoor scene sensing to see if there is a moving or slightly moving human body in the area, and to refresh the detection results in real time.

The maximum sensing distance of a moving human body is 8 m. The sensing distance range, sensing sensitivity in different intervals and refresh time can be easily configured.

GPIO and UART interfaces are plug-and-play and can be flexibly applied to different smart scenarios and terminal products.

1.1. Main Features

- Equipped with single-chip intelligent millimeter wave sensor SoC and intelligent algorithm firmware
- Ultra-small module size: 20 mm × 20 mm
- Load the default human body sensing configuration, plug and play
- 24 GHz ISM band, can pass FCC, CE, and RF spectrum regulations certification
- 3.3 V power supply, supports 3.0 V ~ 3.6 V wide voltage range
- Average operating current 50 mA
- Detection target is moving or slightly moving human body
- Report detection results in real time
- Provides visualization tools to support configuration of detection distance intervals, setting sensitivity within intervals and result reporting time
- Supports sensing range division, completely shielding any interference outside the range

- Short distance sensing of 0.2 m, no blind spot
- Maximum distance of human motion sensing: 8 m
- Large detection angle, coverage range up to ±60°
- Support various installation methods such as ceiling hanging and wall hanging
- Trigger and hold states are configured independently, with strong anti-interference capability

1.2. Application Scenarios

HLK-LD2420 human body sensing sensor can detect and identify moving, standing and stationary human bodies, and is widely used in various IoT scenarios, covering the following types:

Smart Home

Sense the presence and distance of human body and report the detection results for the main control module to intelligently control the operation of home appliances.

Smart Business

Identify when a person is approaching or moving away within the set distance range; light up the screen in time and keep the device lit when a person is present.

Smart Security

Induction access control, building intercom, electronic cat's eye, etc.

Smart Lighting

Identify and sense the human body, accurately detect the position, and can be used for lighting equipment in public places (induction lamps, bulbs, etc.).

2. System Description

HLK-LD2420 is an intelligent and accurate human body sensing sensor developed based on the Hilink S3 series millimeter wave sensor chip.

Use FMCW frequency modulated continuous wave, combined with radar signal processing and built-in intelligent human body sensing algorithm to detect human targets in the set space and update the detection results in real time. Using the Hilink intelligent millimeter wave sensor reference solution, users can quickly develop their own precise human body sensing products.

2.1. Product

The hardware of HLK-LD2420 is mainly composed of the fully integrated Hilink intelligent millimeter wave sensor SoC, 24 GHz one-transmit-one-receive antenna and main control MCU; the software part is equipped with firmware and visual configuration tools released by Hilink, which can flexibly configure the sensing distance, sensitivity and human body sensing function for reporting time.

The specifications of HLK-LD2420 are shown in Table 2-1.

Parameter	Remark	Minimum	Typical	Maximum	Unit
Hardware specifications					
Supported frequency bands	Comply with FCC, CE, and NFPA certification standards	24		24.25	GHz
Supports maximum sweep bandwidth			0.25		GHz
Maximum equivalent isotropic radiated power			11		dBm
Supply voltage		3.0	3.3	3.6	VDC
Size			20 × 20		mm
Ambient temperature		-40		85	°C
System Performance					
Distance detection range (wall mounted)	Sports human target		8		m
	Micro-motion human target		6		m
Distance detection range (ceiling mounted)	Moving human target (diameter)		5		m
	Micro-motion human target (diameter)		4		m
Distance detection accuracy	Moving targets within 8m of the radar straight line distance		±0.35		m
Average operating current			50		mA
Data refresh rate			10		Hz

Table 2-1

3. Hardware Description

The following figure shows the front and back of the module. The module has 5 reserved pin holes (not equipped with pins when leaving the factory) called J2, which is used for power supply and communication; J1 is SWD interface, used for MCU program burning and debugging.

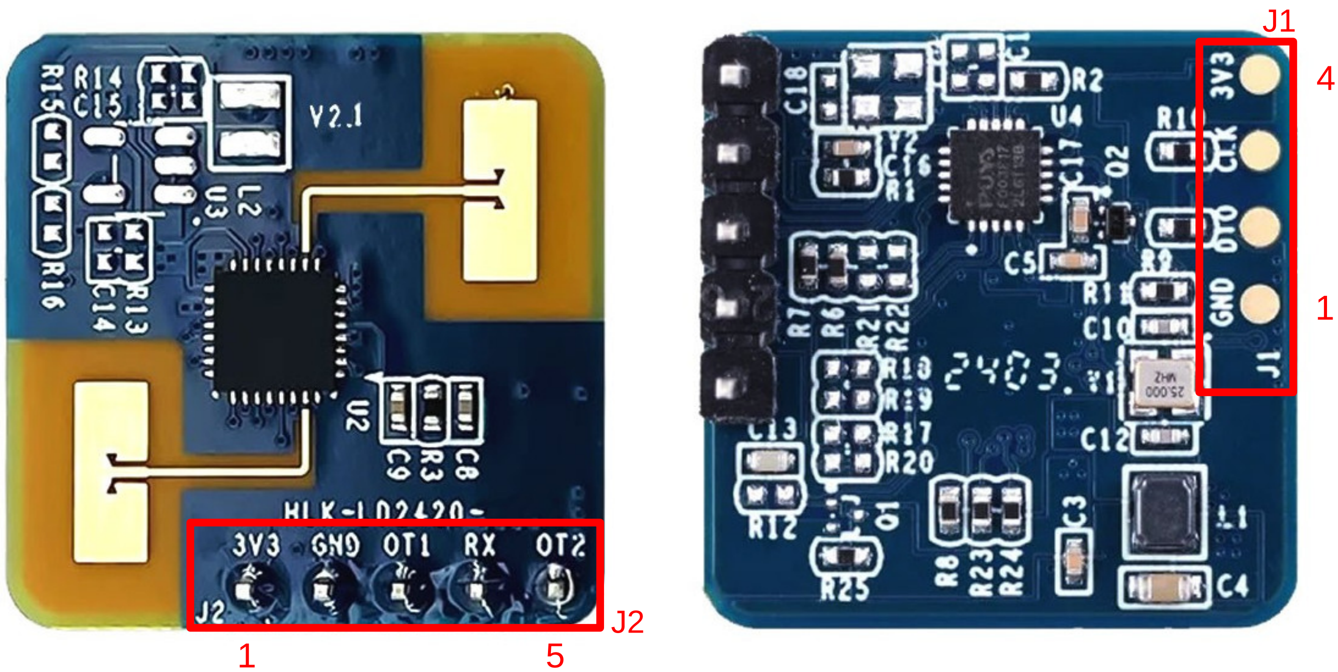


Table 3-1: J1 pin description

J#PIN#	Name	Function	Comment
J1Pin1	GND	Grounding	
J1Pin2	DIO	SWD interface data cable	0 ~ 3.3V
J1Pin3	CLK	SWD interface clock line	0 ~ 3.3V
J1Pin4	3V3	Power Input	3.0 ~ 3.6V, Typ. 3.3V

Table 3-2: J2 pin description

J#PIN#	Name	Function	Comment
J2Pin1	3V3	Power Input	3.0 ~ 3.6V, Typ. 3.3V
J2Pin2	GND	Grounding	
J2Pin3	OT1	UART_TX	0 ~ 3.3V
J2Pin4	RX	UART_RX	0 ~ 3.3V
J2Pin5	OT2	Digital out, used to report detection status: high level means human detected, low level means no detection	0 ~ 3.3V

Note: The pin spacing for J1 and J2 interfaces is 2.54 mm.

4. Software Description

This chapter introduces the firmware debugging of HLK-LD2420 and the use of host computer tools.

HLK-LD2420 has been pre-installed with system firmware. For details on the firmware version, please refer to the module packaging.

The visual host configuration tool software makes it convenient for developers to configure the parameters of HLK-LD2420 according to the usage scenarios and optimize the sensing effect.

4.1. Firmware Configuration

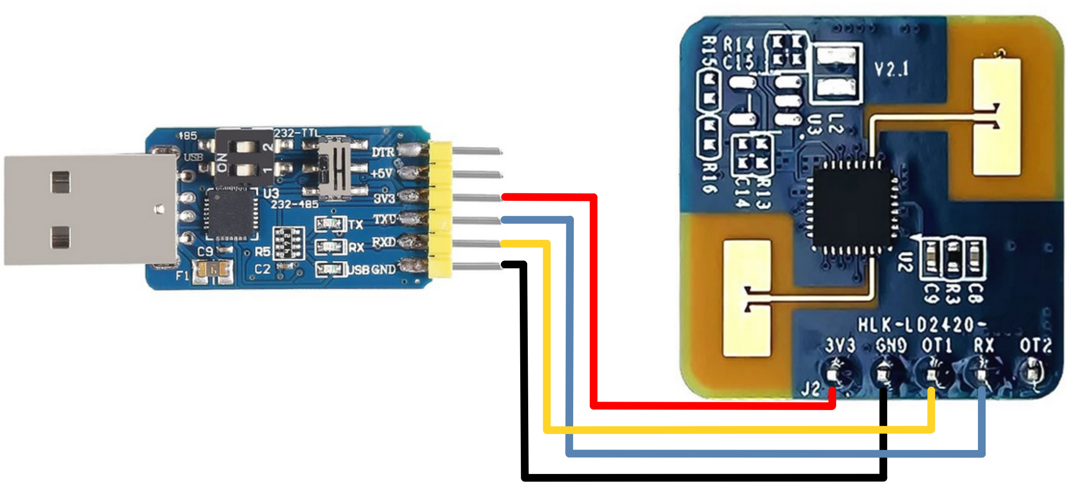
This section introduces the debugging method of HLK-LD2420 radar module firmware.

Step 1: Connect the host computer and radar module through a USB to TTL serial port adapter board like the one below. The pin-to-pin connections are shown in Table 4-1 and Figure 4-1.

Table 4-1: The corresponding relationship between the pins when the radar is connected to the USB serial port adapter board.

Radar Module	Serial Port Adapter Board
RX	TXD
OT1	RXD
3V3	3V3
GND	GND

Figure 4-1: Connection between HLK-LD2420 hardware and USB serial port adapter board



Step 2: Open the device manager of the host computer and check the serial port number of the radar module.

Step 3. Open the serial port tool, select the serial port number of the radar module, set the serial port baud rate to 115200, and then click the "Open Serial Port" button to view the detection results of the current radar at the output end of the tool interface.

4.2. Use of host computer

This section introduces the use of the host computer tool that comes with the HLK-LD2420 module to help users understand the meaning of related parameters and related parameters.

Connection Steps:

Step 1: Get the host computer tool "HLK-2420_TOOL.exe" for HLK-LD2420 from the official website of Hilink.

Step 2: Use the serial port adapter board to connect the radar module and the host computer according to Figure 4-1.

Step 3: Open the host computer tool, select the serial port number of the radar module, enter the baud rate 115200, and click the "Connect Device" button.

Read and write parameters (Note: serial port tools and host computer tools cannot be used at the same time).

4.2.1. Parameter settings

The upper computer tool interface is shown in Figure 4-2.

For detailed explanation of the parameters involved in the host computer tool interface, please see Table 4-2.

Table 4-2: Table 4-2 Parameter explanation of the host computer tool interface

Parameter name	Explanation	Parameter range
Minimum distance	Used to set the minimum distance gate for radar detection. The resolution of the range gate is 70 cm.	0 ~ 15
Maximum distance	Used to set the maximum distance gate for radar detection. The resolution of the range gate is 70 cm.	0 ~ 15 (must be no less than the minimum distance)
Target disappears	The target state needs to be switched from occupied to unoccupied for a period of time T:	(Corresponding to the host computer setting) 0-90

Parameter name	Explanation	Parameter range
	during this period, if a person is detected, the timer for this period will be restarted.	
Delay time	The state of no one will switch to no one after it lasts for a full T time. Human status, report no one.	(Corresponding to serial port configuration) 0 ~ 1000000000
Trigger threshold	Used to set the sensitivity from no-man to manned state. It is recommended to set it at 5 times the energy. For energy scanning and data viewing, please refer to 4.2.2 and 4.2.3.	(Corresponding to the host computer setting) 0-90 (Corresponding to serial port configuration) 0 ~ 1000000000
Keep Threshold	The sensitivity used to detect human micro-movement and maintain the status of people is recommended to be set at 2 ~ 5 times the background noise. For energy scanning and data viewing, please refer to 4.2.2 and 4.2.3	(Corresponding to the host computer setting) 0-90 (Corresponding to serial port configuration) 0 ~ 1000000000

Threshold parameter description: Assume that N is the parameter configured by the host computer and M is the parameter configured by the serial port.

The parameter conversion relationship is $N = (10 * \log_{10} M)$ or $M = 10^{N/10}$.

For example, if the serial port configuration distance gate 0 threshold value is 65536, in the host computer it should be $(10 * \log_{10} 65536) \approx 48.16$.

Or, if the host computer parameter is set to 70, the corresponding serial port configuration parameters are 10 70/10 = 10,000,000 and the instruction converts to hexadecimal, with little endian, to: 0x80969800

4.2.2. Real-time data

The "Real-time Data" page of the host computer is shown in Figure 4-4. Its function page is introduced as follows:

- The colored light icon in the upper left corner indicates whether there is someone in the detection area: when the radar detects the presence of a human body, the colored light is red; when no human body is detected; when present, the colored light is green;
- The text display box behind the colored light shows the radial distance between the target human body detected by the radar and the radar;
- The "Start/Pause" switch button is used to start and pause radar detection;

- The "Generate Threshold" button is used to scan the ambient noise and calculate the "trigger threshold" and "hold threshold" of each range gate;
- The "Apply Threshold" button is used to send the "Trigger Threshold" and "Hold Threshold" obtained in the "Generate Threshold" function to the radar;
- The "Relative Power VS Range Gate" line graph is used to display the motion energy value (green line), trigger threshold value (red line) and hold threshold value (yellow line) of each range gate in real time; a black background indicates that the range gate is in the effective detection range, and a gray background indicates that the range gate is in the invalid detection range;
- The "Distance vs. Time" line graph is used to display the distance change of the target human body detected by the radar in the past 60 seconds in real time; the gray background area indicates that the radar detected the target human body during this period, and the black background area indicates that the radar did not detect the target human body during this period.

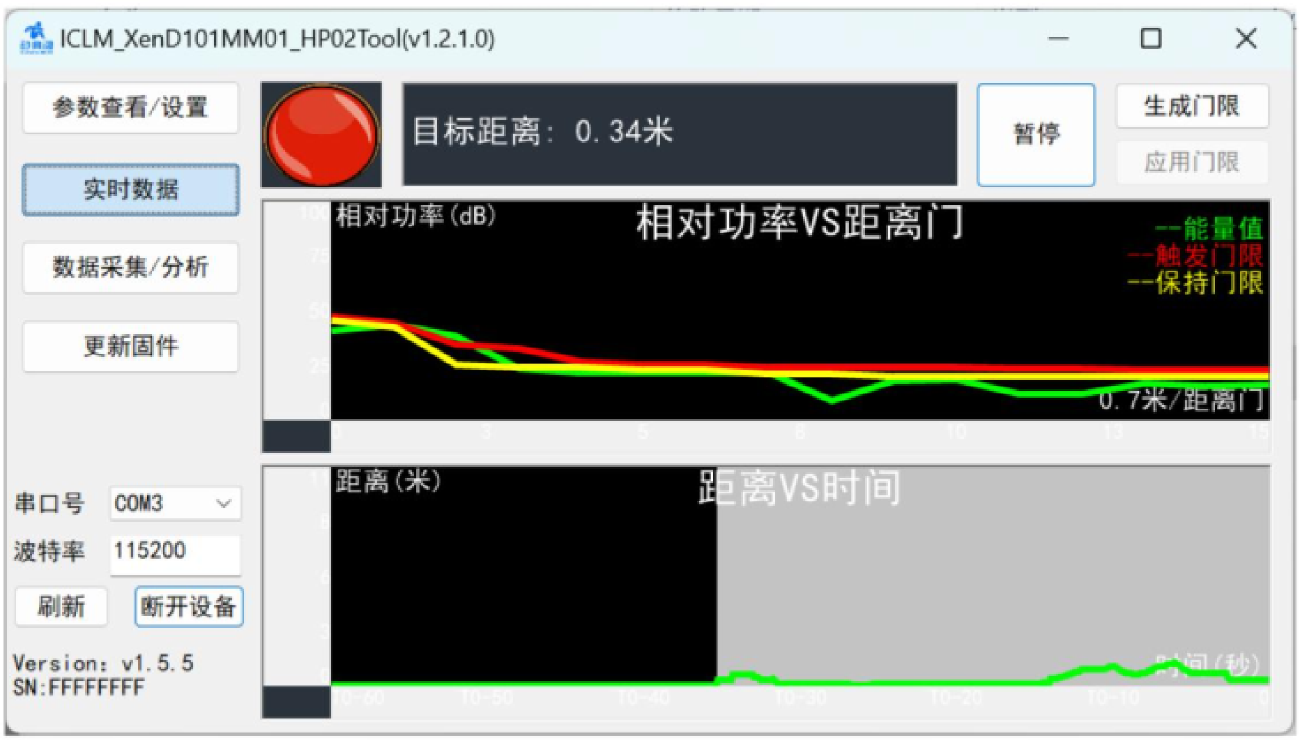


Figure 4-4 Real-time data page

The steps to view real-time data through the host computer are as follows:

Step 1: After connecting the XenD101MM module to the host computer tool, click the "Real-time Data" button to switch to this function page.

The tool automatically turns on the radar detection function, and the "Start/Pause" switch button displays "Pause". The two line graphs on the upper computer function page start to display the corresponding real-time data information;

Step 2: Click the "Start/Pause" switch button to pause the radar detection function. The colored light on the function page turns green and the target distance displays "0.00 Meters", the two line charts below stop updating.

The steps to generate/apply thresholds via the host computer are as follows:

Step 1: When the "Start/Pause" button on the "Real-time Data" page shows "Pause", click the "Generate Threshold" button and the host computer tool pops up the "Threshold Acquisition" information window. The table above shows the trigger energy and hold energy values of each distance gate in real time, and the progress bar below displays the scanning progress, as shown in Figure 4-5;

Step 2: Click the "Cancel" button to terminate the collection; if the user wants to save and apply the collected data, click "OK" after the data collection is completed;

Step 3 (optional): If the user selects "OK" in step 2, the "Apply Threshold" button on the page changes from a gray, unclickable state to a clickable state. Click the "Apply Threshold" button, and the host computer tool sends the threshold data calculated in step 2 to the radar, and a prompt window appears saying “Threshold setting successful” pops up.

After applying the generated threshold, the user can view the latest radar threshold parameter values on the "Parameter View/Set" page.

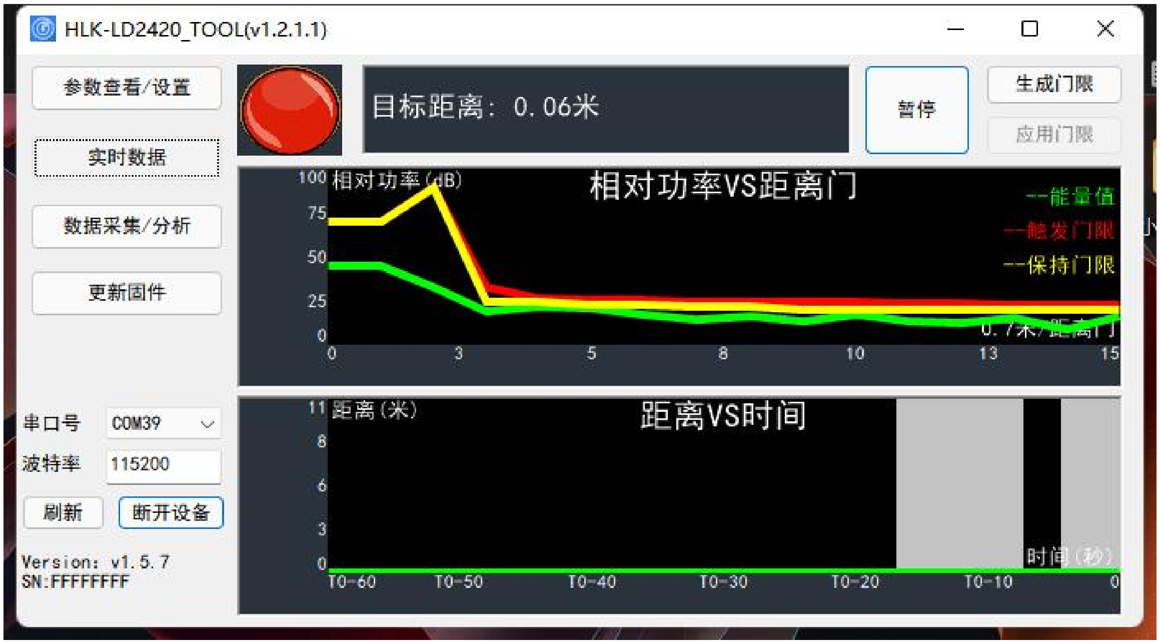
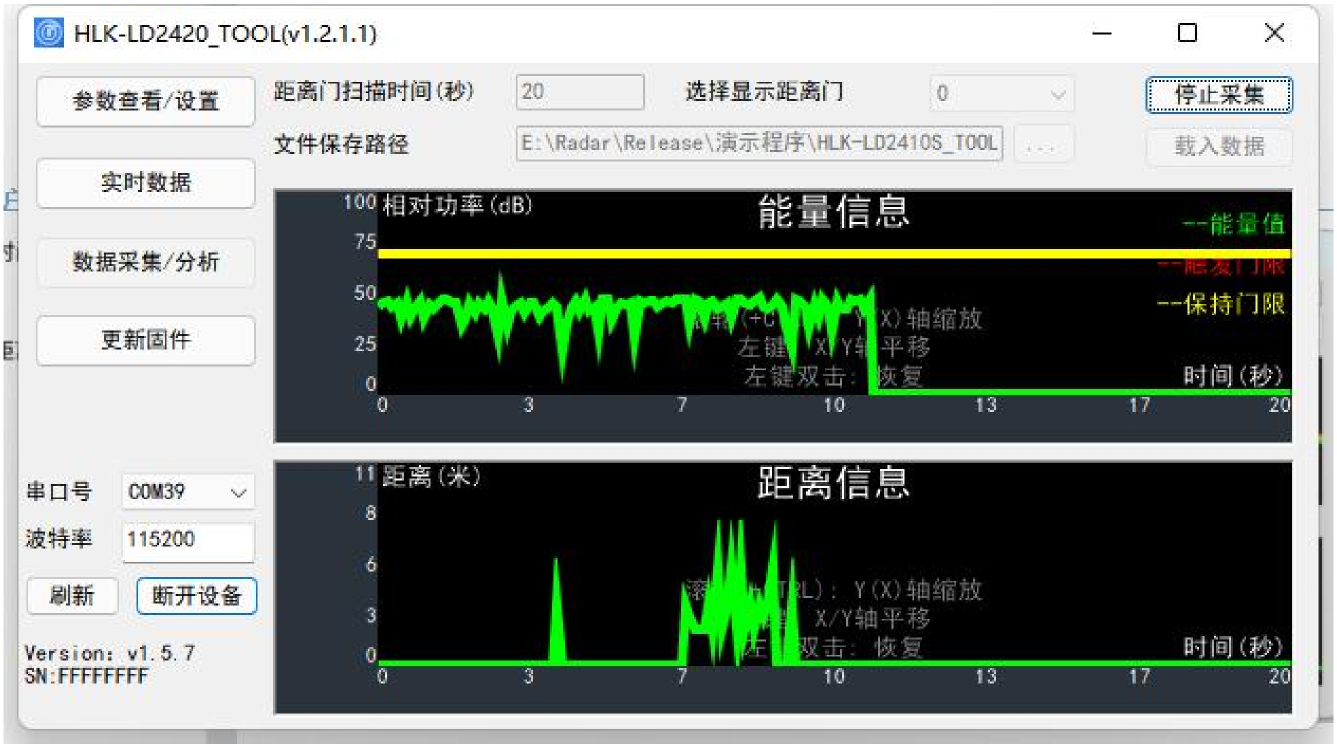


Figure 4-5 Threshold Collection Page

4.2.3. Data collection/analysis

The "Data Collection/Analysis" page of the host computer is shown in Figure 4-6. Its function page is introduced as follows:

- "Range gate scanning time (seconds)": used to set the environmental noise scanning time of each range gate, the default is 20 seconds, the value range is 0 ~ 65535;
- "File save path": used to set the save path of the collected data;
- "Select display range gate": used to select the range gate to be viewed, the optional range is 0 ~ 15;
- "Collect Data/Stop Collection" switch button: used to start and stop data collection. After stopping data collection, users can save the data in the set file.
- Under the path, you can see a .dat file with a file name starting with RadarData and ending with a timestamp;
- “Load Data” button: used to open the saved radar scan data for users to view and analyze;
- “Energy Information” line graph: used to display the sweep energy value, trigger threshold, and hold threshold on the range gate selected by the user. The horizontal axis is time. The vertical axis is the energy information expressed as relative power;
- “Distance Information” line chart: used to display the distance information of human targets detected within the radar detection range, with the horizontal axis representing time and the vertical axis representing distance.



4-6 Data Collection/Analysis Page

The steps to collect energy data through the host computer are as follows:

Step 1: After connecting the XenD101MM module to the host computer tool, click the "Data Collection/Analysis" function button to switch to the function page;

Step 2: Enter the "range gate scanning time", set the "file saving path":, make sure there is no one in the radar detection area within a scanning cycle, and click the "Collect Data/Stop Collecting" switch button to start collecting data;

Step 3: After starting to collect data, the user can wait for the host computer tool to automatically stop collecting after the scan is completed, or click "Collect Data/Stop" button to stop data collection in advance; in both cases, the collected data will be saved in the file save path set in step 2.

Step 4: If you want to view the specific data of a certain point on the curve, place the mouse cursor at the position of interest on the curve, and a floating frame will appear next to the cursor. The energy value or distance information at that location is displayed, as shown in Figure 4-7.

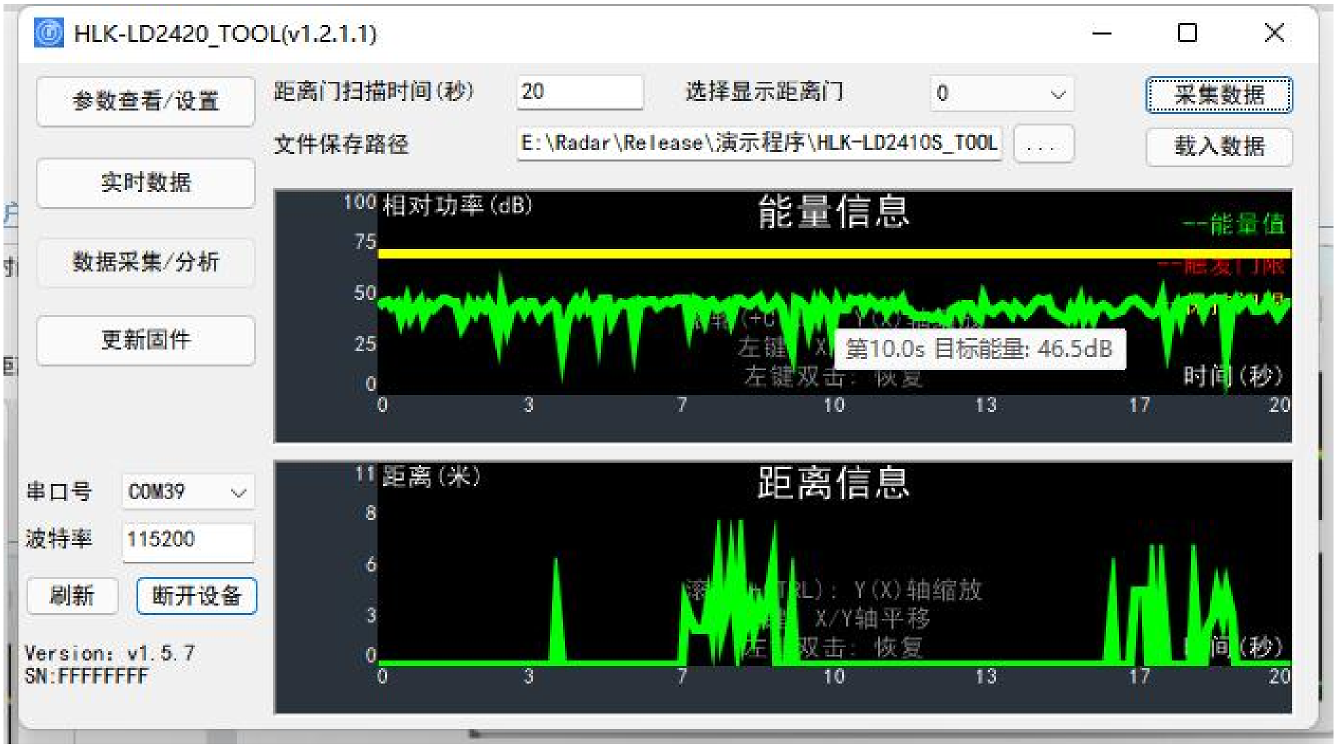


Figure 4-7 Data viewing page

When viewing data, users can zoom, pan, and restore the line chart:

- Zoom the horizontal coordinate: Put the mouse cursor on the graphics canvas, press the Ctrl key on the keyboard and roll the mouse wheel to zoom in (roll the mouse wheel up) or zoom out (roll the mouse wheel down) the graphics on the horizontal coordinate;

- Shift the data graph: put the mouse cursor on the graph canvas, press the left mouse button and move the mouse to move the graph accordingly, and the horizontal and vertical coordinates will also change synchronously;

- Restore graphics: Place the mouse cursor on the graphics canvas and double-click the left mouse button to restore the graphics to the default display settings.

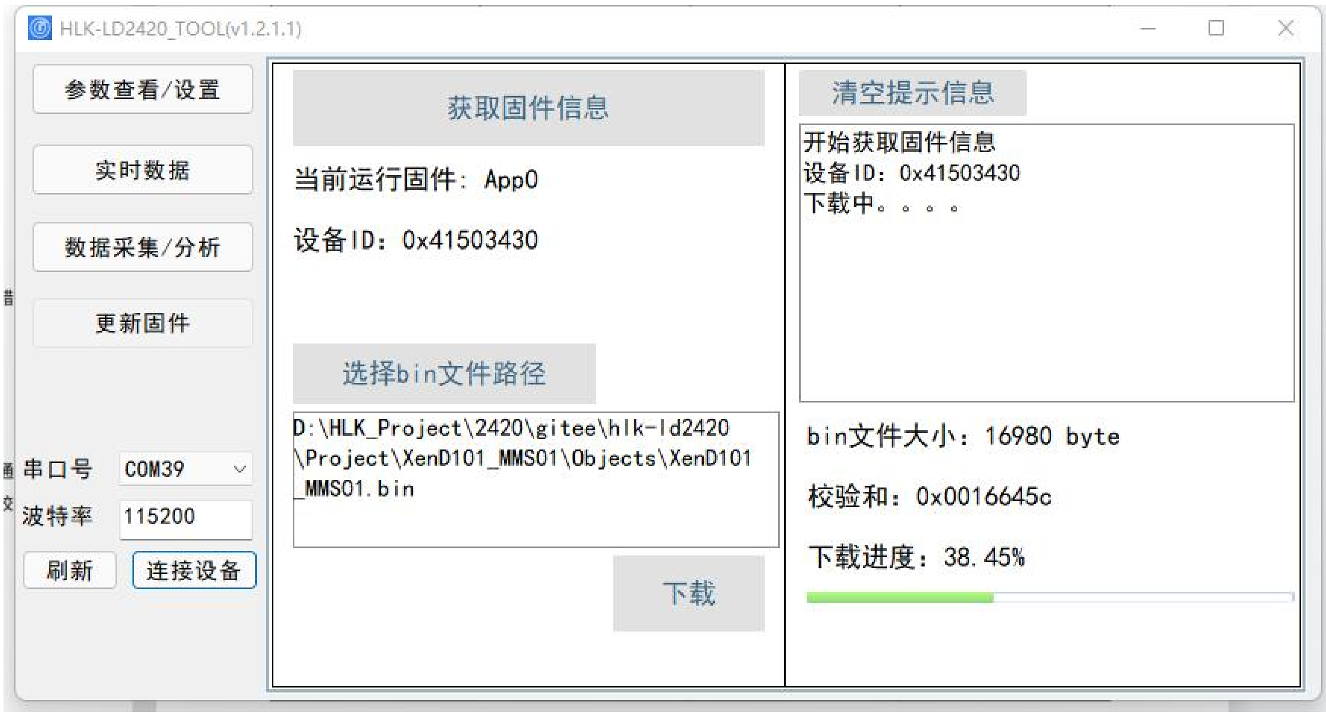
4.2.4. Update Firmware

The "Update Firmware" page of the host computer is shown in Figure 4-8. The steps to update the radar module firmware through the host computer are as follows:

Step 1: After connecting the XenD101MM module to the host computer tool, click the "Update Firmware" function button to switch to the function page;

Step 2[3]. Click the "Get Firmware Information" button on the function page, and the ID information of the current device will be displayed in the prompt box on the right;

Step 3. Click the "Select bin file path" button, select the required .bin file, and click the "Download" button to start upgrading the firmware. The information box will display the download results in real time, and the bin file information and current download progress will be displayed below.



4-8 Firmware Upgrade Page

After the firmware upgrade is successful, the page prompt box will display "Download Successful!". If the firmware upgrade fails, the prompt box will display the corresponding error message information.

3 This step is required and cannot be skipped when the user updates the firmware using the host computer interface.

5. Communication Protocol

This communication protocol is mainly used by users who need to conduct secondary development without visualization tools. HLK-LD2420 communicates with the outside world through the serial port (TTL level). The radar's data output and parameter configuration commands are all carried out under this protocol. The default baud rate of the radar serial port is 115600, 1 stop bit, and no parity bit.

This chapter mainly introduces this communication protocol from three parts:

- Protocol format: including protocol data format and command frame format;
- Configure command package format: including command package format and command return package format;
- Upload data frame format: including the upload data frame format of the debugging mode and the upload data frame format of the reporting mode.

The basic process of using commands to configure parameters is:

1. Enter command mode;
2. Configuration parameter command/get parameter command;
3. Exit command mode.

5.1. Protocol Format

5.1.1. Protocol data format

HLK-LD2420 data communication uses the little-endian format. All data in the following table are in hexadecimal.

5.1.2. Command protocol frame format

The radar configuration command and ACK command formats defined by the protocol are shown in Table 5-1 and Table 5-3.

Table 5-1 Send command protocol frame format

Frame Head	In-frame Data Length	In-frame Data	Frame Tail
FD FC FB FA	2 bytes	See Table 5-2	04 03 02 01

Table 5-2 Data format in the sending frame

Command word (2 bytes)	Command value (N bytes)

Table 5-3 ACK command protocol frame format

Frame Head	In-frame Data Length	In-frame Data	Frame Tail
FD FC FB FA	2 bytes	See Table 5-4	04 03 02 01

Table 5-4 ACK frame data format

Command word (2 bytes)	Command execution status (2 bytes)	Return value (N bytes)

5.2. Send command and ACK

5.2.1. Read firmware version command

This command reads the radar firmware version information.

Command word: 0x0000

Command value: None

Return value: Version number length (2 bytes) + version number byte string

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9 ~ 12
FD FC FB FA	02 00	00 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11, 12	Byte 13 ~ 18	Byte 19 ~ 24
FD FC FB FA	0C 00	00 01	00 00	06 00	76 31 2E 35 2E 25	04 03 02 01

5.2.2. Enable Configuration Command

Any other command issued to the radar must be executed after this command is issued, otherwise it will be invalid.

Command word: 0x00FF

Command value: 0Xx0001

Return value: 2 bytes ACK status (0 success, 1 failure) + 2 bytes protocol version (0x0002) + 2 bytes buffer size (0x0020)

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	FF 00	01 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11, 12	Byte 13, 14	Byte 15 ~ 18
FD FC FB FA	08 00	FF 01	00 00	02 00	20 00	04 03 02 01

5.2.3. End configuration command

End the configuration command. After execution, the radar resumes working mode. If you need to send other commands again, you need to send the enable configuration command first.

Command word: 0x00FE

Command value: None

Return value: 2 bytes ACK status (0 success, 1 failure)

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9 ~ 12
FD FC FB FA	02 00	FE 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	FE 01	00 00	04 03 02 01

5.2.4. Read Serial Number Command

This command reads the radar's serial number.

Command word: 0x0011

Command value: None

Return value: 2 bytes ACK status (0 success, 1 failure)

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9 ~ 12
FD FC FB FA	02 00	11 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11, 12	Byte 13, 14	Byte 15 ~ 18
FD FC FB FA	08 00	11 01	00 00	02 00	CD AB	04 03 02 01

5.2.5. Write Serial Number Command

This command writes the radar's serial number.

Command word: 0x0010

Command value: SN length (2 bytes) + SN byte string

Return value: 2 bytes ACK status (0 success, 1 failure)

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9 ~ 12	Byte 13 ~ 16
FD FC FB FA	06 00	10 00	02 00 CD AB	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	10 01	00 00	04 03 02 01

5.2.6. Read radar register command

This command reads the radar's registers.

Command word: 0x0002

Command value: 2-byte chip address + (2-byte address) * N

Return value: (2 bytes of data) * N

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9 ~ 12	Byte 13 ~ 16
FD FC FB FA	06 00	02 00	40 00 40 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11, 12	Byte 13 ~ 16
FD FC FB FA	06 00	02 01	00 00	07 02	04 03 02 01

5.2.7. Configure radar register commands

This command writes to the radar's registers.

Command word: 0x0001

Command value: 2-byte chip address + (2-byte address) * N

Return value: 2 bytes ACK status (0 success, 1 failure)

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14	Byte 15 ~ 18
FD FC FB FA	08 00	01 00	40 00	40 00 07 42	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	01 01	00 00	04 03 02 01

5.2.8. Read radar parameter configuration command

This command can read the current configuration parameters of the radar.

Command word: 0x0008

Command value: (2 bytes parameter ID) * N

Return value: (4 bytes parameter value) * N

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	08 00	01 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14	Byte 15 ~ 18
FD FC FB FA	08 00	08 01	00 00	0C 00 00 00	04 03 02 01

5.2.9. Configure radar parameters command

This command sets the parameters of the radar module. For specific parameter words, please refer to Table 5-5.

Table 5-5 Protocol parameter table

Parameter name	Parameter word	Parameter range
Minimum distance gate	0x0000	0 ~ 15
Maximum distance gate	0x0001	0 ~ 15
Target disappearance delay	0x0004	0 ~ 65535 Unit: seconds
Trigger threshold	0x0010 ~ 0x001F	0 ~ 232 - 1, square of modulus value
Hold Threshold	0x0020 ~ 0x002F	0 ~ 232 - 1, square of modulus value

Command word: 0x0007

Command value: (2 bytes parameter ID + 4 bytes parameter value) * N

Return value: 2 bytes ACK status (0 success, 1 failure)

Send data: Maximum distance gate 12, Minimum distance gate 2

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11, 12	Byte 13, 14	Byte 15 ~ 18
FD FC FB FA	08 00	07 00	01 00	12 00	00 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	07 01	00 00	04 03 02 01

5.2.10. Configuration system parameter commands

This command can configure radar system parameters.

Command word: 0x0012

Command value: (2-byte parameter ID + 4-byte parameter value) * N

Return value: 2 bytes ACK status (0 success, 1 failure)

Send data:

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14	Byte 15 ~ 18
FD FC FB FA	08 00	12 00	00 00	04 00 00 00	04 03 02 01

Radar ACK (success):

Byte 1 ~ 4	Byte 5, 6	Byte 7, 8	Byte 9, 10	Byte 11 ~ 14
FD FC FB FA	04 00	12 01	00 00	04 03 02 01

5.3. Radar reporting data

HLK-LD2420 firmware outputs the test results through the serial port in normal working mode. The output result is the string of ON/OFF and the target distance gate.

In special modes, the host machine can obtain data from the radar processing process, so the firmware provides two additional transmission formats in command line mode:

They are debug mode and report mode respectively.

In command line mode, by adjusting the working mode parameters in the command packet, the data format reported by the serial port can be controlled. Figure 5-1 shows a command packet example.

FD FC FB FA	08 00	12 00	00 00	04 00 00 00	04 03 02 01
Frame header	Command part length	Command ID	Parameter ID	Parameter value	Frame tail

Command Parameter

5-1 Command Packet Format Example

In this command packet, the parameter value of the command parameter can determine the working mode of the radar: 0x0000 is the debugging mode, 0x0004 is the reporting mode, and 0x0064 is the normal mode.

Table 5-6 shows the data format of the debugging mode. The RDMAP modulus value data transmitted in the debugging mode is sent according to the data of all range gates of each chirp, that is, the modulus value data of the 16 range gates of the first chirp is sent first, and then the modulus value data of the 16 range gates of the second chirp is sent, and so on.

Table 5-6 Data frame format in debug mode

Frame Head	Data	Frame Tail
AA BF 10 14	RDMAP: 20 (Doppler) * 16 (range gate) * 4 (squared modulus)	FD FC FB FA

Table 5-7 Data frame format of reporting mode

Header	Length	Detection result	Target distance	Energy value of each distance gate	Frame tail
F4 F3 F2 F1	2 bytes, total number of bytes of detection result, target distance and energy value of each distance gate	1 byte, 00: no one 01: someone	2 bytes	32 bytes 16 (total number of distance gates) * 2 bytes	F8 F7 F6 F5

Figure 5-2 shows an example of a data frame transmitted from the serial port in reporting mode.

Header Length Result Distance

F4 F3 F2 F1 23 00 01 69 00 (F2 1B 6A 07 14 00 28 00 1D 00 19 00 14 00 20 00 0D 00 32 00 19 00 14 00 0A 00 0A 00 08 00 0A 00) F8 F7 F6 F5

Energy value of each distance gate Tail

Figure 5-2: Data frame example in reporting mode

In the example, the meaning of each part of the data is as follows:

Length field: indicates the total number of bytes occupied by the reported detection result + target distance + energy value of each range gate;

Detection result: Indicates whether there is a target in the current scene;

Target distance: indicates the distance between the target in the scene and the radar;

Energy value of each range gate: Indicates the energy value of each range gate from 0 to 15.

Therefore, the data frame analysis example in Figure 2 is as follows:

Length: 0x0023, indicating the total number of bytes occupied by the reported detection result + target distance + energy value of each range gate = 35

Detection result: 01, indicating that there is a person in the scene

Target distance: 0x0069, means target distance = 105 cm

Energy value of each range gate:

Range Gate Energies			
Gate 0	0x1BF2 : 7154_{10}	Gate 8	0x000D : 13_{10}
Gate 1	0x076A : 1898_{10}	Gate 9	0x0032 : 50_{10}
Gate 2	0x0014 : 20_{10}	Gate 10	0x0019 : 25_{10}
Gate 3	0x0028 : 40_{10}	Gate 11	0x0014 : 20_{10}
Gate 4	0x001D : 29_{10}	Gate 12	0x000A : 10_{10}
Gate 5	0x0019 : 25_{10}	Gate 13	0x000A : 10_{10}
Gate 6	0x0014 : 20_{10}	Gate 14	0x0008 : 8_{10}
Gate 7	0x0020 : 32_{10}	Gate 15	0x000A : 10_{10}

6. Installation and detection range

HLK-LD2420 supports two installation methods: ceiling-mounted and wall-mounted. The recommended method is ceiling-mounted installation.

The orientation of the radar is shown in Figure 5-1. The Y axis is 0°, the X axis is 90°, and the Z axis is perpendicular to the XY plane (also called the line direction).

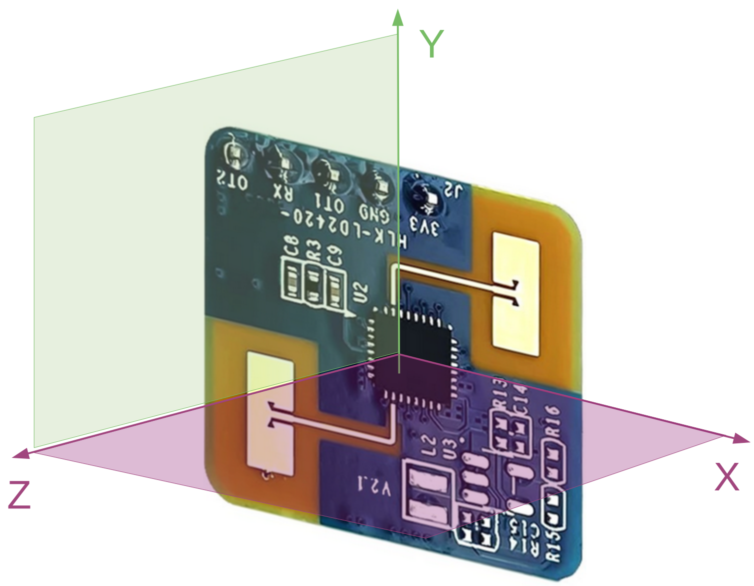


Figure 5-1 Radar module orientation diagram

6.1. Ceiling installation

The recommended ceiling installation height is 2.7~3 m. The maximum motion sensing range of the ceiling-mounted HLK-LD2420 radar module under the default configuration is a conical three-dimensional space with a bottom radius of 5m is shown in Figure 5-2.

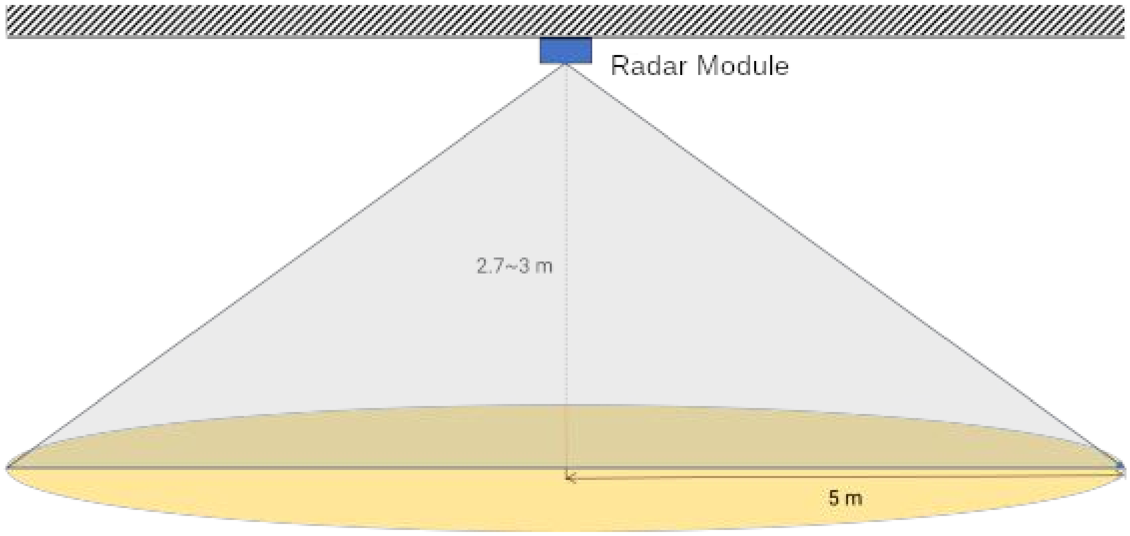
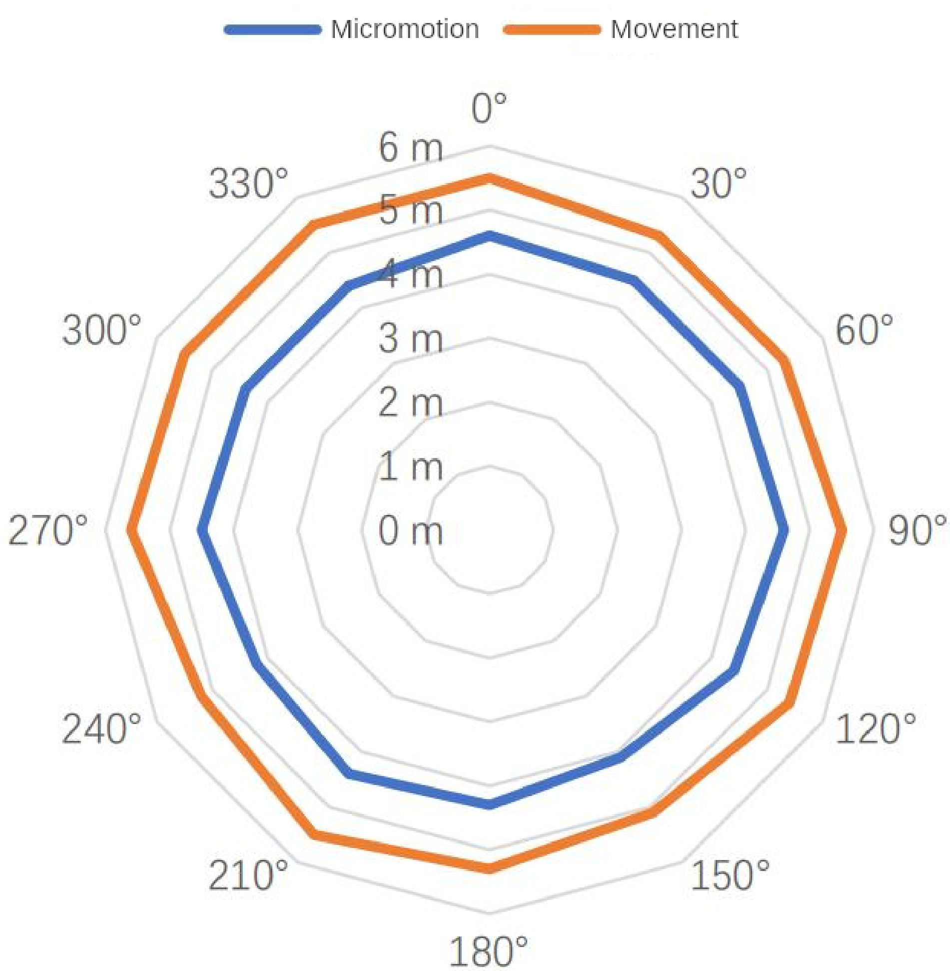


Figure 5-2 Schematic diagram of the detection range of the HLK-LD2420 radar module (ceiling mounted)

The schematic diagram of the motion and micro-motion detection range of this reference solution when the ceiling installation height is 2.7 m is shown in Figure 5-3.



6.2. Wall Mounting

The recommended wall-mounted installation height is 1.5~2 m. When mounted on the wall, the X-axis of the radar module (see Figure 5-1) points horizontally and the Y-axis points upward.

The Z axis points to the detection area. The maximum motion sensing range of the wall-mounted HLK-LD2420 radar module in the default configuration is 8 m in radius and horizontal

A three-dimensional fan-shaped space with an angle of ±60° with the pitch direction, as shown in Figure 5-4.

The detection range diagram of this reference solution when the wall-mounted installation height is 1.5m is shown in Figure 5-5.

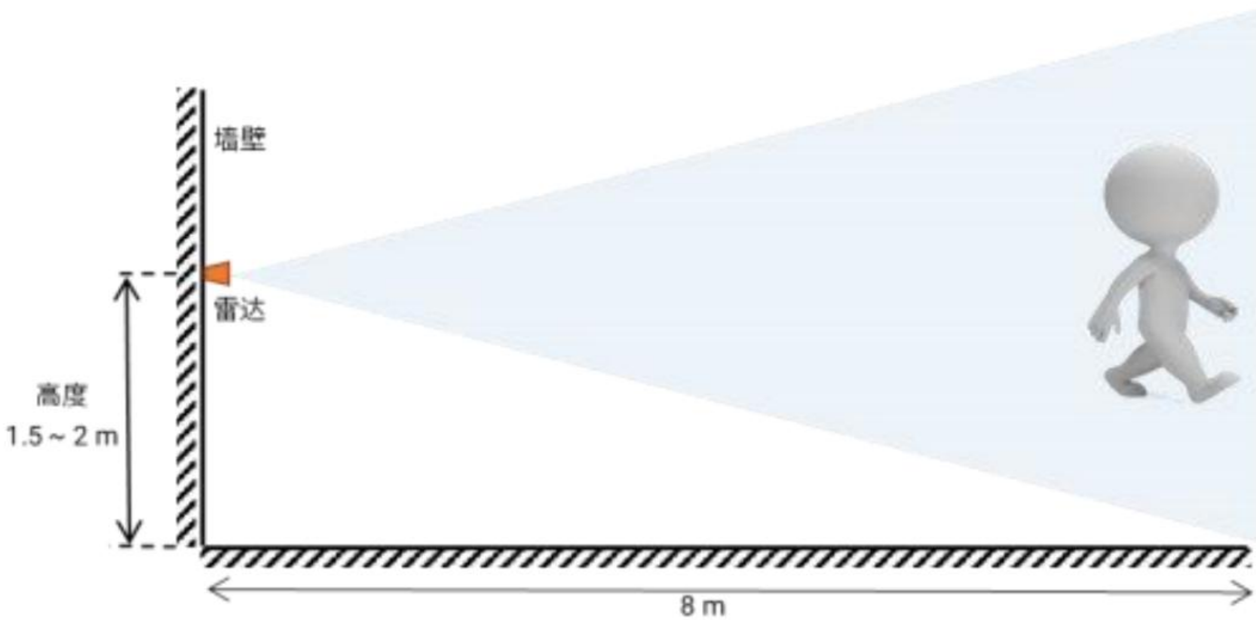


Figure 5-4 Schematic diagram of the detection range of the HLK-LD2420 radar module (mounted on the wall)

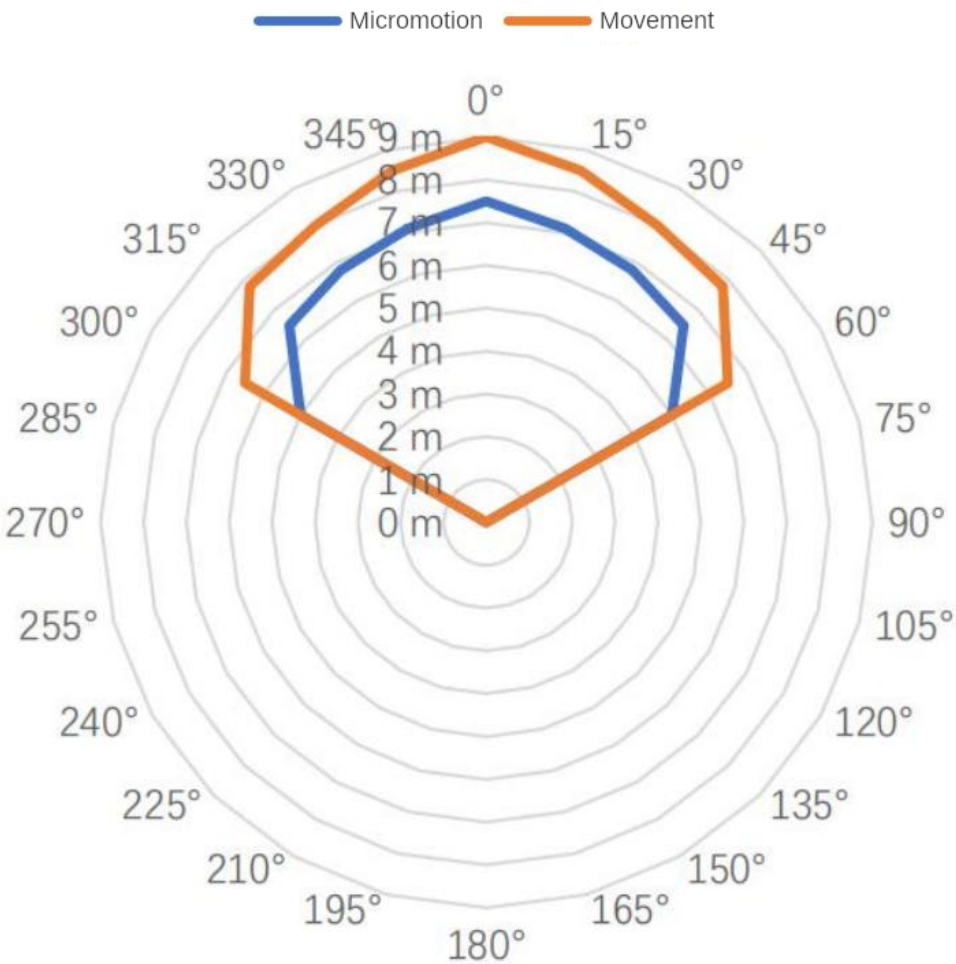


Figure 5-5 Wall-mounted sensing range

6.3. Detection range test

The test methods for radar triggering and maintaining detection range are introduced as follows:

- Trigger range:

The target person approaches the radar from a distance when the radar reports no person. When the radar starts to report that there is a person, it stops moving forward. The current position is the boundary of the radar trigger detection range; the area enclosed by the detection boundaries in all directions is the radar trigger detection range.

- Keep range:

When the target person keeps a small movement at the position to be detected, such as shrugging or raising his hand, the radar will detect the person within 60 seconds.

If there is someone reported all the time, the current position is within the radar detection range; otherwise, the detection position is outside the detection range.

7. Mechanical dimensions

Figure 6-1 shows the mechanical dimensions of the module, all units are in millimeters. The module board thickness is 1.2 mm, with a tolerance of ±10%.

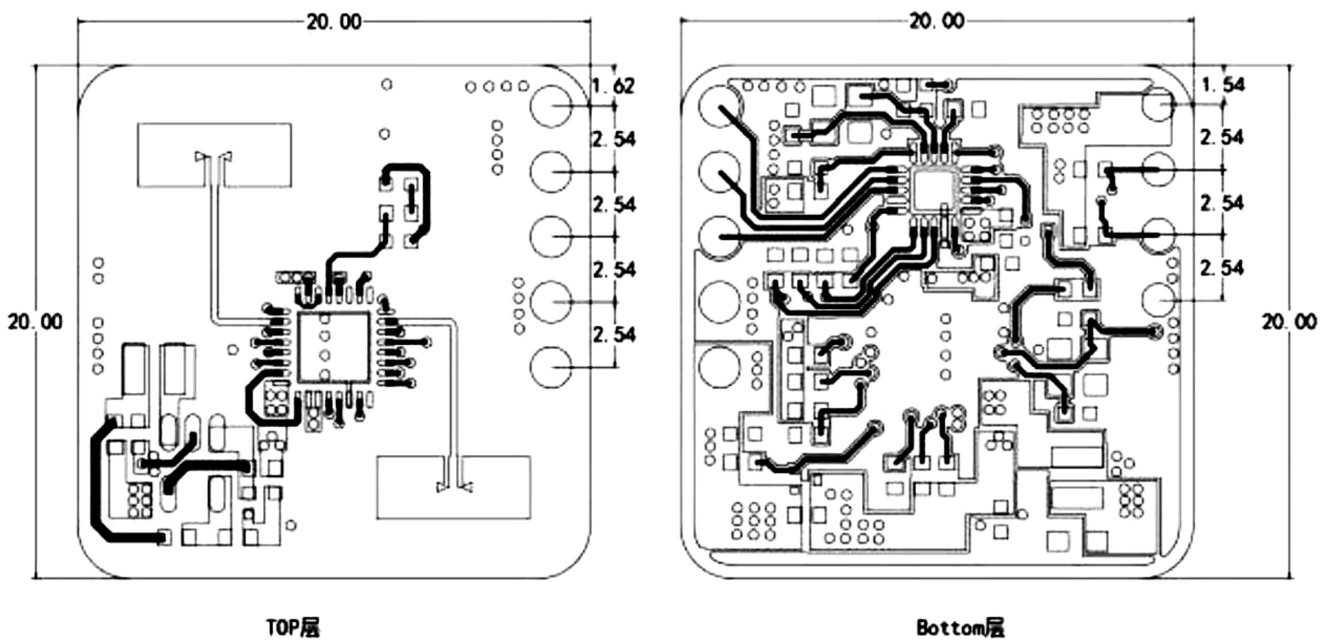


Figure 6-1 Hardware mechanical dimensions

8. Installation Instructions

Radar enclosure requirements

If the radar needs to be installed in a housing, the housing must have good wave-transmitting properties in the 24 GHz band and must not contain metal or electromagnetic wave shielding.

For more information, see the mmWave Sensor Radome Design Guide.

Installation environment requirements

This product needs to be installed in a suitable environment. If used in the following environment, the detection effectiveness will be affected:

- There are non-human objects that are in continuous motion in the sensing area, such as animals, continuously swinging curtains, and large green plants facing the air vents.

- There is a large area of strong reflective plane in the sensing area. Strong reflective objects facing the radar antenna will cause interference.
- When wall-mounted, external interference factors such as air conditioners and electric fans on the ceiling of the room need to be considered.

Notes on installation

- Try to ensure that the radar antenna is facing the area to be detected and that there is an open area around the antenna.
- Ensure that the sensor is installed firmly and stably. The shaking of the radar itself will affect the detection effect.
- Ensure that there is no movement or vibration of objects behind the radar. Since radar waves are penetrating, the antenna back lobe may detect moving objects behind the radar. Metal shielding covers or metal back plates can be used to shield the radar back lobe and reduce the impact of objects behind the radar.
- When there are multiple 24 GHz frequency band radars, please avoid direct beam alignment and install them as far away as possible to avoid possible mutual interference.

Power Supply Considerations

- The power input voltage range is 3.0 V~3.6 V, and the power ripple has no obvious spectrum peak within 100 kHz. This solution is a reference design.
- Users need to consider the corresponding electromagnetic compatibility design such as ESD and lightning surge.

9. Notes

Maximum detection distance

The maximum range of the radar to detect a target is 8 m in radial distance. Within the detection range, the radar will report the straight-line distance of the target from the radar.

It can only output the distance information of moving human bodies within 8 m, and does not currently support the close-range distance measurement function of stationary human bodies.

Maximum distance and accuracy

Theoretically, the radar ranging accuracy of this reference solution is 0.35 m. Due to the different body shapes, states, and RCS of human targets, the ranging accuracy will fluctuate, and the maximum detection distance will also fluctuate to a certain extent.

Target disappearance delay

When the radar module detects that there is no human presence in the target area, it will not immediately report the "no one" status in the area, but will be delayed.

The mechanism of delayed reporting is: once no human target is detected within the test range, the radar module will start timing, and the duration is the duration of no one. If no one is detected during the timing, the "no one" status will be reported after the timing ends; if a person is detected during this time period, the timing will be immediately ended and updated, and the target information will be reported.

10. Version History

Version	Date	Changes
V1.0	2023/2/15	Initial draft.
V1.1	2023/3/23	The pin function of the module has changed, the default baud rate has changed to 115200, and the host computer version has been updated
V1.2	2023/8/15	Added serial port description, new version host computer usage instructions, serial port command protocol
V1.3	2023/12/11	Modify parameter description
V1.3.1	2024/3/21	Modify parameter description
V1.3.2	2024/11/22	Modify parameter description